

CPL Empirical Study: Final Revision Pass

Cross-Project Bug Localisation: What Works, What Doesn't, and Why. An Empirical Investigation | A. Eashaan Rao, S. Chimalakonda

Core message. The study seems solid, but the writing has to be revised. Most earlier items are fixed; a few errors remain that a careful referee will catch, plus a proposed summary box for the front of the paper. Nothing below needs new experiments. The core RQ1 result and the controlled multi-instrument protocol are clean.

Proposed summary box (place at end of the Introduction)

A boxed synthesis on page 1 or 2 delivers what the title promises and surfaces the calibration insight that is otherwise only in Section 5.2. Render it in the house style (thin black border, white background, Liberation Sans). Once it is in, trim the Section 4.4 synthesis so the paper does not state the same thing three times.

Summary of Findings

What works. Cross-project transfer outperforms within-project limited training for 2 of 3 architectures: embedding-based dual-encoders (BLAZE, $d = 1.19$, win rate 90.5%) and CNN-based rerankers (TRANP-CNN, $d = 0.59$, win rate 68.3%). Embedding-based models deploy with zero target annotations, statistically equivalent to limited within-project training (TOST, plus/minus 0.05 MRR). CNN-based rerankers with 20% target labels reach the full within-project ceiling (TOST, plus/minus 0.05 MRR).

What doesn't. The GNN-based adversarial model (COOBA) shows only marginal gains ($d = 0.27$, no metric surviving Bonferroni correction). Cold-start fails for both scalar-ranking models. Negative transfer affects roughly one in three deployments for TRANP-CNN (27%) and COOBA (32%), against 3.2% for BLAZE. No project-level feature (codebase size, domain gap, source quality) reliably predicts pair-level success for any model.

Why. Transfer viability appears to depend on whether a model's ranking mechanism requires target-side calibration at inference time. Embedding-similarity ranking carries cross-project semantics in its representation and transfers without target labels; learned-scalar ranking is calibrated to the source and cannot rank a new target without them. This single axis is consistent with the cold-start dichotomy, the negative-transfer spectrum, and the cross-model disagreement.

Key insight. CPL outcomes are architecture-dependent, not just a property of the source-to-target pair. Cross-model agreement is 41 to 67%: the same pair can help one architecture and harm another. A CPL result therefore does not generalise from one tool to another, and practitioners should validate empirically for their chosen model.

Must fix

- 1. Conclusion paragraph.** "Our findings includes Cross-project transfer significantly outperforms..." is a grammar break in the last-read paragraph. Rewrite it properly, restore the RQ1/RQ2/RQ3 structure, and add back the CNN-based deployment pathway, which the closing currently drops (only embedding-based tools are mentioned).
- 2. Zenodo DOI.** Still reads PLACEHOLDER in Data Availability. Archive the package and insert the real DOI. This is a desk-check trigger.
- 3. Power claim (Section 3.9).** ">80% power to detect a small effect ($d = 0.2$)" is wrong; the true value at $n = 63$ for a one-sided paired test is near 50 to 55%. Recompute, or drop the small-effect clause and claim only the medium-effect power. This matters because COOBA sits at $d = 0.27$ and the paper leans on being adequately powered.
- 4. Over-strong sentence (Section 4.1.2).** Cut "A large effect ($d > 1$) means the CPL gain is detectable even in studies with as few as 10 project pairs: the benefit is unmistakable in scale." The extrapolation is disputable and "unmistakable" overclaims. The effect sizes and CIs already carry it.

Should fix

- 5. Pseudo-replication.** Thirteen targets recur across 63 pairs (about five times each), so the pairs are not independent, and the variance paragraph makes this visible by noting twelve WP-small runs for jupyterlab. Add two sentences in conclusion validity naming non-independence, noting the median-WP-small re-test partially addresses it, and leaving a mixed-effects model (target as random effect) to future work. Owning it converts a likely criticism into a disclosed limitation.
- 6. TOST margin justification (Section 3.9).** Deriving the equivalence margin from the observed effect (TRANP-CNN's +0.054) is circular. Reground the 0.05 MRR margin in the triage/rank-shift argument already

used for the 0.20 viability threshold.

7. **Case 3 inconsistency (Section 4.5).** The Case 3 table shows TRANP-CNN WPL = 0.138 for scikit-learn, but Table 8 gives large-target TRANP-CNN WPL = 0.242, and 0.138 for 80% training is implausibly low. Verify the value; if it is a genuine outlier, add a clause.
8. **Table 7 footnote.** COOBA MAP is starred significant while its d CI crosses zero. Legitimate (parametric CI, non-parametric test), but add a one-line footnote so it does not read as an inconsistency.

Overclaims to soften

9. **"Removing the annotation cost entirely."** Given the leakage caveat now disclosed, soften to "removing the need for target annotations in these deployments" in the abstract and Section 5.1.
10. **"Strongest observable moderator" (Section 4.2.1)** sits awkwardly against "no single feature reliably predicts" in 4.2.2. Change to "the clearest size-stratified moderator" to avoid the apparent contradiction.
11. **Keep the hedges.** "Appears to govern" (5.2) and "appears to be a stronger determinant" (4.3) are correct; do not clean them back to "governs" or "determines."

Writing pass

12. **Qualifiers.** The test is whether the word states something checkable or tells the reader how to feel. Keep the first, cut the second. Keep "pronounced (27x)," "large" with d and CI, "steepest (28pp)," "marginal," "near-symmetric," "smallest": all quantified or descriptive. Cut "notable finding," "notable improvements," "notable degradation" (use the 14.5-point figure instead), "particularly striking," "reveals a structured pattern," "depends critically." Do not over-correct into blandness; where a number sits in the same sentence, an adjective pointing at it is fine.
13. **"Significant."** Keep only where a test sits beside it. Scan for any remaining loose uses.
14. **Citation thinning.** Section 2 and the Section 3 pair-selection sentence carry three-to-seven-citation strings that read as padding. The generic methodology triple (Wohlin, Runeson, Feldt) stapled to a compute-budget sentence is the clearest case. Keep the citations that carry a claim, remove the rest.

Figure and framing

15. **Figure 1 label touch-ups.** The end-to-end figure is good and needs no rework, only: "Beetlebox" to "BeetleBox," "TRANP CNN" to "TRANP-CNN," the RQ1 box "WPL vs CPL" to "CPT vs WPS" (RQ1 is CP-transfer vs WP-small), "Mods" to "Moderators," and add "+ size/domain diversity" to the funnel so it matches the caption.
16. **Surface the insight in the results.** The calibration mechanism is the paper's central idea but appears only in Section 5.2. Add one bridging sentence in the Section 4.4 synthesis stating it plainly as the explanation for architecture-dependence, so the insight is findable in the results, not only in the discussion.

JSS-specific flags, with suggested wording

For each item the suggested text is a starting point. Read it, check it against the data, and adjust before using.

17. **Format.** Stay in the Elsevier CAS double-column template; it is the JSS default and the figures and wide tables are already sized for it. Single column has no reviewer upside here and would mean re-fitting every figure. If a single-column review copy is wanted, use elsarticle with the review option, then switch back for the final. No text change needed.

Current: (double-column CAS template, as now)

Suggested: (keep as is)

18. **"State-of-the-art" is a dating risk.** TRANP-CNN (2019) and COOBA (2021) are not the current frontier, and the 2025 references now in the paper (SweRank, FunLoc, the knowledge-enhanced localiser) make that visible. Reword the roughly four occurrences (abstract, intro, conclusion, 3.2). You chose these models because they are architecturally representative, so say that.

Current: "three state-of-the-art bug localisation models"

Suggested: "three architecturally distinct, well-established CPL models spanning the dominant paradigms (embedding-based, convolutional, and adversarial GNN)"

19. **COOBA re-implementation fidelity.** The paper concedes re-implementation but implies fidelity without evidencing it, and COOBA is the exposed case because it is the weak model. Add a sentence to the re-implementation threat. If any anchor to the original's behaviour exists, add it; if not, concede the limit plainly.

Current: "The evaluated systems are therefore faithful architectural adaptations rather than exact replicas..."

Suggested: "The evaluated systems are faithful architectural adaptations rather than exact replicas. We cannot fully exclude the possibility that COOBA's weak CPL profile reflects our re-implementation rather than the adversarial paradigm itself, though the shared retrieval stage and identical protocol control for the largest sources of variation. A closer replication of each original model is left to future work."

- 20. Compounded selection effect.** Small corpus, popular-repo bias, and pretraining exposure are handled as three separate caveats but are one compounded selection effect. Add a sentence to external validity acknowledging the interaction.

Current: (the three caveats appear separately across the threats section)

Suggested: "These three constraints are not independent: the corpus is small, drawn from benchmarks that favour popular well-instrumented repositories, and those same repositories are the most likely to appear in the embedding backbone's pre-training. The combined selection effect should be weighed when generalising the absolute performance figures, though the within-model CPL comparisons are less exposed to it."

- 21. Pairs-over-seeds tradeoff.** The disclosed variance invites the question of why the compute went to more non-independent pairs rather than more independent seeds. Pre-empt it with one sentence, near the run-to-run variance paragraph.

Current: (not currently addressed)

Suggested: "The compute budget was directed toward breadth of source-to-target pairs rather than repeated seeds per pair, since the cross-model consistency question (RQ3) requires many pairs evaluated identically across models. This favours statistical power for the architecture-level claims at the cost of per-pair variance estimates, which the median re-test above partially recovers."

- 22. Section 4.4 opening (framing alignment).** The synthesis still opens with the old utility framing, now out of step with the reframed abstract and the new summary box.

Current: "Cross-project learning is a viable strategy for data-scarce bug localisation in 2 of 3 architectural paradigms tested, but its effectiveness and risk profile are architecture-dependent."

Suggested: "Across three architectures and 63 pairs, CPL outcomes are architecture-dependent rather than a property of the source-to-target pair. Transfer is beneficial for 2 of 3 paradigms, but its effectiveness and risk profile differ sharply by architecture."

- 23. Consistency nits.** Copyedit-level, but a reviewer notices them.

Current: "source-to-target" and "source-arrow-target" alternate; "GNN-based adversarial models (COOBA)" is plural for one model.

Suggested: Pick one of source-to-target / source->target throughout. Use the singular where one model is meant: "the GNN-based adversarial model (COOBA)". Same for BLAZE and TRANP-CNN where pluralised.